CYM Case 2432S028R – 3D Printing and Installation Manual

Case with battery and GPS for the ESP32-2432S028R (CYD-2432S028, 2.8" touch screen) · Revision 1

Designed and made by @birolt29 @alen29





Real device (CYM menu open) and 3D design view (front).

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This manual has four parts: general information about the board and the parts used, 3D printing of the case and the front frame, assembling the parts into the case, and installing the firmware. Print settings are given by their generic names; you will find the equivalents in whichever slicer you use.



Finished device: front, SD / charge wall, USB wall, speaker wall.



Design views: front, back, inside of the case, USB wall.

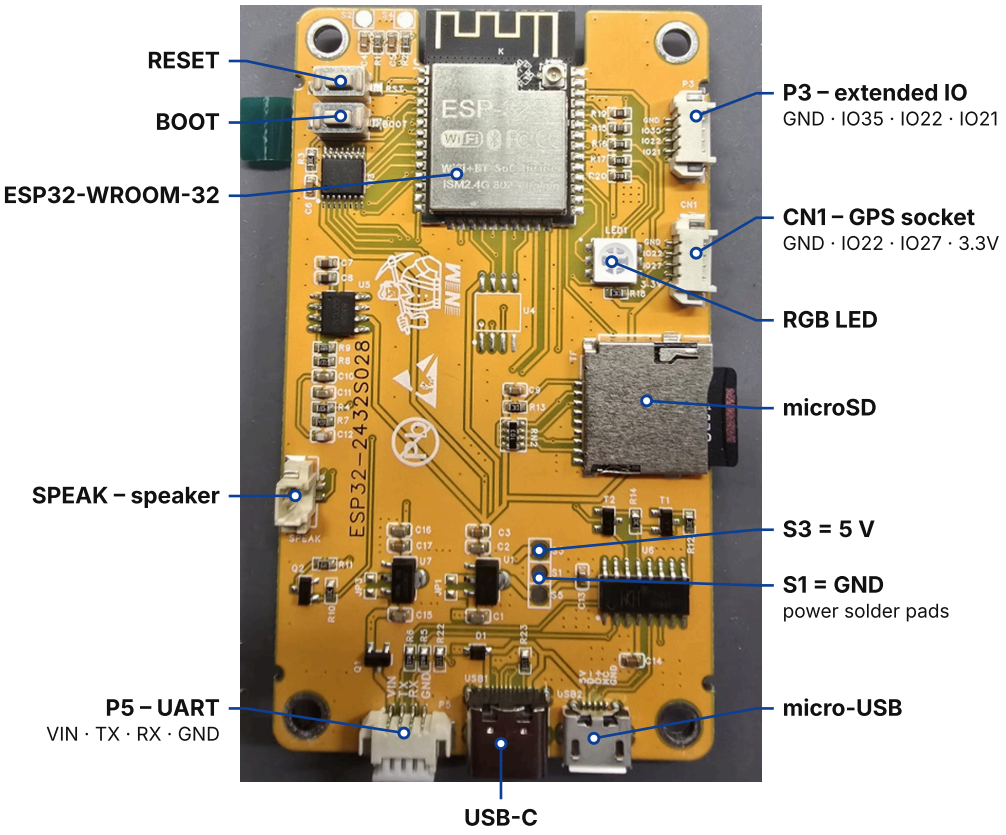
Part 1 – General information

Board specifications

Processor	ESP32-WROOM-32 · dual core · 240 MHz
Memory	4 MB Flash
Wi-Fi	802.11 b/g/n · 2.4 GHz
Bluetooth	Bluetooth 4.2 (Classic + BLE)
Display	2.8" TFT · 320×240 px · ILI9341 driver
Touch	Resistive · XPT2046 driver · stylus included
RGB LED	RGB LED (on board)
Speaker	Speaker output (SPEAK socket)
microSD	SPI interface · FAT32 · max 32 GB

USB	USB-C + micro-USB
Power	5 V via USB, or battery + charge module (this case)
Case dimensions	88.9 × 52.9 × 15.8 mm (case) + 3.5 mm (front frame) = 19.3 mm total

Connectors (pinout)



Back of the board; USB sockets at the bottom.

CN1 – GPS socket (4-pin, 1.25 mm)

Pin	Signal	Description
1	GND	Ground
2	IO22	Board RX – GPS TX connects here
3	IO27	Board TX – GPS RX connects here
4	3.3V	3.3 V power output (GPS supply)

S1 / S3 – power solder pads

Pin	Signal	Description
S1	GND	Power minus
S3	5 V	Power plus (from the charge module)

Board connectors: CN1 GPS socket (GND–IO22–IO27–3.3V) and S1/S3 power solder pads.

Reference: parts used and software

Part	Product	Link
Board	ESP32-2432S028R (CYD-2432S028), ESP32 with 2.8" touch screen	www.amazon.de/dp/B0D2TN77LY
Battery	3.7 V 1100 mAh LiPo with a 1.25 mm plug. Sold under the "YELUFT" name; the product arrives labeled "MakerFocus".	www.amazon.de/dp/B0FX8M8DK7

GPS module	ATGM336H GPS+BDS Dual Mode Module	www.amazon.de/dp/B0FKH26344
Charge module	Youmile LX-LCBST charge + 5 V boost module (Type-C), pack of 5	www.amazon.de/dp/B0C3V847G2
Switch	SS12D00G slide switch, 3 mm lever ("12D00G-3mm"); from the VEXUNGA 80-piece slide-switch set	www.amazon.de/dp/B0FG2TSLFC
Connector cable (optional)	Huba 1.25 mm 2-pin male + female connector cable pairs, 10 cm, 20 pairs	www.amazon.de/dp/B0GRGJBH13
Software	Cheap Yellow Monster (CYM)	github.com/JimGat/CYM#supported-hardware

The board comes as a kit with its screws, standoff nuts, stylus and 4-wire cable; you need to buy the battery, the charge module, the GPS module and the switch separately. The case was tested with these parts. Before ordering, compare the parts' dimensions with the list above: the battery pocket in the case is 53.4 × 31 × 7.5 mm, and batteries of the same capacity can vary in size from maker to maker. Do not trust listed dimensions either: the battery used here is listed as 50 × 29 × 7 mm, but it actually measures 53 × 30 × 7.5 mm (including the protection board and the wire exit). Links were checked on 2026-10-04; product pages may change over time. The software supports this board under the name CYD-2432S028; select it as the board on the install page.

> Warning – battery polarity: on batteries with a 1.25 mm plug and on ready-made connector cables, the plus/minus order varies by maker, and the wire color is not reliable either. Before connecting the battery to the module, use a multimeter to confirm which wire is positive. Reversed polarity will destroy the charge module.

Part 2 – 3D printing

Print files

File	Contents
`CYM_Case_2432S028R.3mf`	Both parts together (case + front frame), in print orientation. Use this if your slicer supports 3MF.
`CYM_Case_2432S028R_Case.stl`	Case (lower body). Units: millimeters.
`CYM_Case_2432S028R_Front_Frame.stl`	Front frame. Units: millimeters.

Two parts are printed; no supports are needed. Case 88.9 × 52.9 × 15.8 mm, front frame 88.9 × 52.9 × 3.5 mm; combined thickness 19.3 mm. Total volume 17.7 cm³ (about 20 g of PLA).

Placement on the build plate

- Case: the flat, closed back face (the one with the BOOT / RESET / GPS lettering) goes on the build plate; the open side faces up.
- Front frame: the flat face that faces the board goes on the build plate; the front face with the CYM lettering faces up.
- The files are saved in this orientation; do not rotate them.
- Supports off. No supports are needed anywhere.
- No brim is needed; add a 3–5 mm brim if the corners lift.

Material

PLA or PLA+ is sufficient. Take the temperatures from your filament's label; since the part prints slowly, the low to middle part of the recommended range works well. Damp filament causes stringing and fills in small lettering; dry it before printing.

Slicer settings (0.4 mm nozzle)

Setting	Value	Why
Layer height	0.16 mm (first layer 0.20 mm)	For small lettering and holes. 0.20 mm also works, but the result is coarser.
Wall (perimeter) count	3	Side walls are 1.2 mm thick; 3 perimeters make them solid.
Wall generator	Arachne / variable line width	Thin ribs and button tabs print as one piece.
Wall printing order	Outer wall first	Outer surface and lettering edges come out crisp.
Outer wall speed	30 mm/s	For the engraved lettering and the button slots.
Flow rate	Calibrated; avoid over-extrusion	Over-extrusion closes the 0.6 mm button slots and the lettering.
Top / bottom solid layers	4 / 4	
Infill	15%	
Elephant foot compensation	0.10 mm	Keeps first-layer squish from narrowing the lettering and slots on the back face.
Supports	Off	

Setting names differ between slicers (Cura, PrusaSlicer, OrcaSlicer, Bambu Studio, Creality Print, etc.). Reference print: an Ender-3 S1-class printer, textured PEI plate, 0.4 mm nozzle, PLA+, nozzle 210 / 205 °C, bed 60 / 55 °C; both parts together take about 2 hours 25 minutes.

Checks after printing

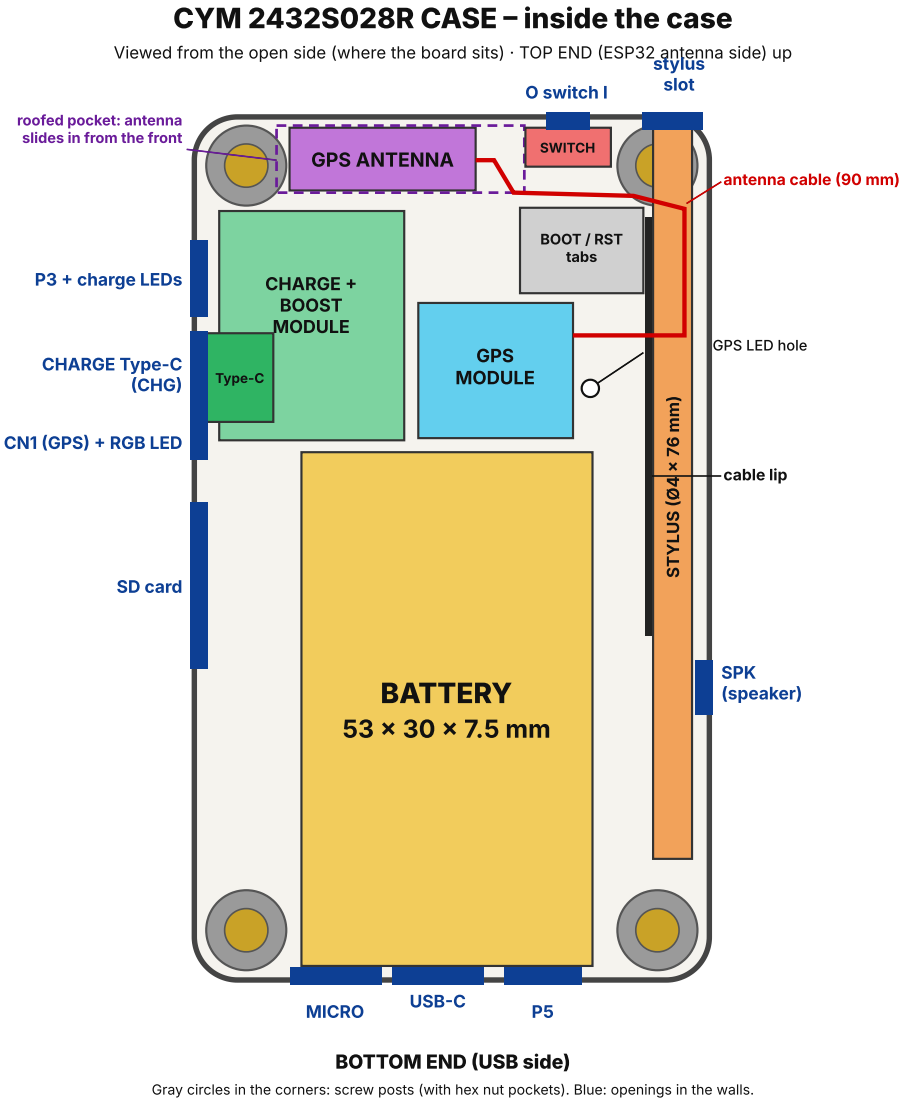
- Do the BOOT and RESET tabs flex freely? If the slits along the tab edges are closed, open them with a hobby knife; the thin slit on the outer edge of the RESET tab in particular can close up in the first layer.
- Are there loose strands from bridging on the ceiling of the GPS antenna pocket? If so, remove them with tweezers before fitting the antenna.
- Are the hex nut pockets in the four posts clean?
- Is the stylus hole open end to end? The stylus must slide in and out easily.
- Are the USB (MICRO / USB-C), P5, P3, CN1, SD, SPK and CHG openings clean?
- Are the 1.5 mm LED holes open (the RGB LED and the two charge LEDs in the side wall, the GPS LED in the back face)? If not, open them by hand with a 1.5 mm drill bit.
- Is there any stringing in the switch slot or inside the engraved lettering?

Part 3 – Assembly

What you need

- Printed case and front frame (CYM Case 2432S028R, revision 1)
- ESP32-2432S028R (CYD-2432S028) board and the parts from its kit: 4 M3 hex standoff nuts, 4 short M3 screws (M3 × 7), the stylus and a 4-wire cable (one end has a plug). The acrylic plates and long screws in the kit are not used.
- Protected 1100 mAh LiPo battery, max. 53 × 30.5 × 7.5 mm, with a 1.25 mm plug
- Charge + 5 V boost module LX-LCBST (23.5 × 19 × 4.5 mm, Type-C input)
- GPS module 16 × 14 × 5.5 mm and its antenna (19 × 6 × 5.5 mm, 90 mm cable)
- Slide switch SS12D00G, 3 mm lever (body 8.7 × 3.7 × 3.7 mm). The same body also comes with 4, 5 and 6 mm levers; a longer lever sticks out farther from the case.
- Thin hook-up wire (for power and the switch)
- Optional: 2-pin 1.25 mm connector pairs (to make the battery and board connections detachable)
- Thin double-sided tape, soldering equipment, multimeter, tweezers, hobby knife

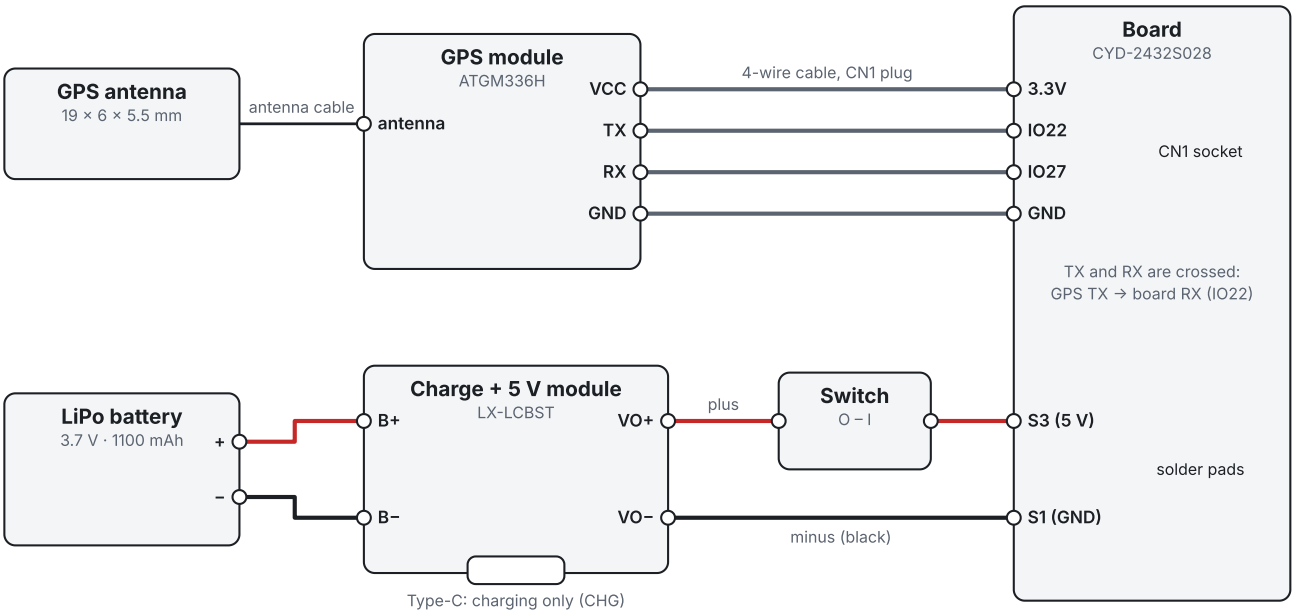
Interior layout diagram



Component layout inside the case (open-side view, top end up).

Wiring diagram

Which wire goes where: the battery connects to the charge module, the module's 5 V output goes through the switch to the board, and the GPS module connects to the CN1 socket.



Keep the switch off (O) while a USB cable is plugged into the board.

> Caution: the wire colors of the GPS module cable and the power wires can differ from set to set; red = + and black = − is a common convention, but do not rely on it. Make the connections by the socket/pad labels, not by wire color (GPS VCC, TX, RX, GND ↔ CN1 3.3V, IO22, IO27, GND; TX and RX are crossed), and verify the battery's positive wire with a multimeter.

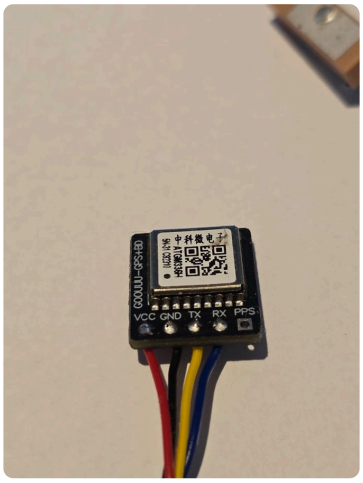
Preparation: soldering work

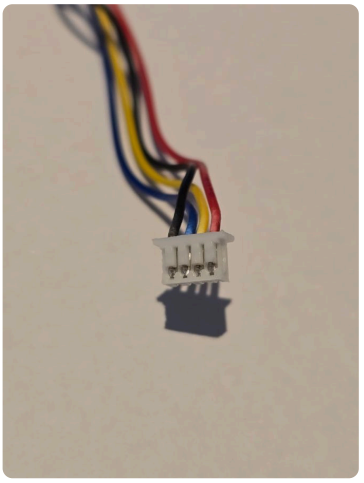
These four tasks are done on the bench, before anything goes into the case.

Preparation 1 Solder the cable to the GPS module

The 4-wire cable from the kit has a plug on one end that fits the CN1 socket; keep that end. Cut the connector off the other end and solder the four wires directly to the GPS module's pads (VCC, GND, TX, RX); there is no room for a pin header. Route the wires flat along the module.

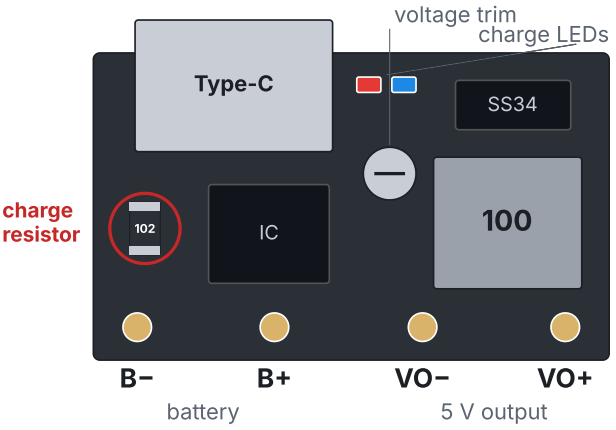
The labels next to CN1 on the board read (top to bottom) GND, IO22, IO27, 3.3V. GPS TX goes to the board's RX (IO22), GPS RX to the board's TX (IO27); VCC → 3.3V, GND → GND. The GPS module runs on 3.3 V.



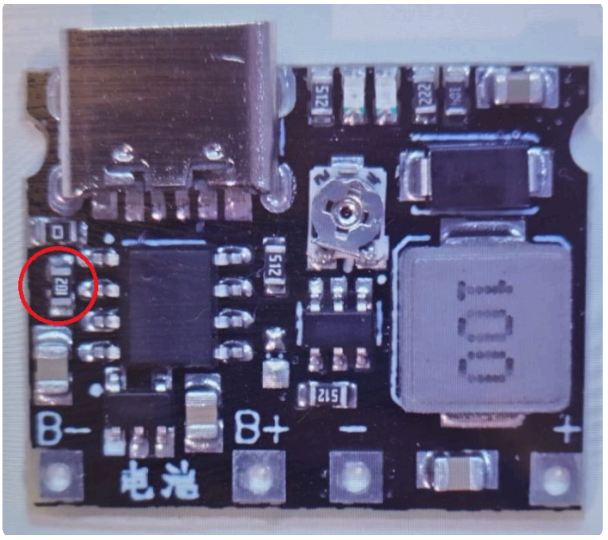


Preparation 2 Set up the charge module

Set the charge module's output voltage to 5 V DC: with the battery connected but the module not yet connected to the board, measure across the VO+ and VO− terminals with a multimeter and turn the trimmer screw with a fine screwdriver until it reads 5.0 V. Do not connect the module to the board until the voltage is set correctly. Set the charge current to suit your battery using the 0603-size resistor on the module. Manufacturer's table: 30 kΩ → 50 mA, 5 kΩ → 250 mA, 2 kΩ → 580 mA, 1.2 kΩ → 1000 mA. The module ships with a 1 kΩ resistor (marked 102), which gives a charge current of about 1 A, the upper limit for the 1100 mAh battery (the battery's recommended charge current is 0.2 – 1 A). Replacing it with a 2 kΩ resistor (580 mA) is recommended.



The drawing is schematic and not to scale. The resistor sits between the Type-C socket and the B− pad, at the edge of the module (marked in red in the photo).



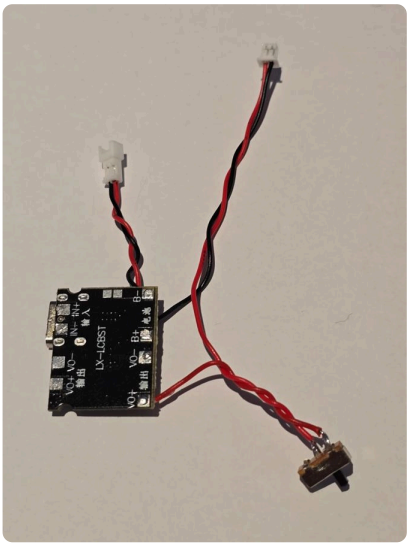
Charge current – resistor

Resistor	Charge current
30 kΩ	50 mA
5 kΩ	250 mA
2 kΩ	580 mA · recommended
1.2 kΩ	1000 mA
1 kΩ ("102")	≈1.2 A · factory

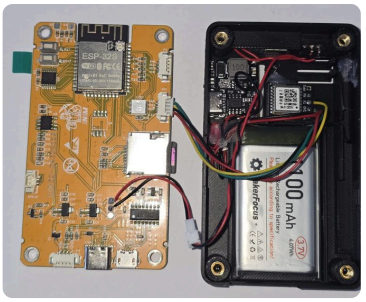
Preparation 3 Wire the charge module and the switch

Solder the battery wires to the B+ and B− pads and the output wires to the VO+ and VO− pads, on the top side of the module. Identify the pads by the markings on the module itself; the arrows in seller images can be misleading. Route the positive output wire through the switch: VO+ → the switch's middle leg, one outer leg → the positive wire to the board.

> Trim the switch legs as short as possible; the legs and the soldered wires must not obstruct the movement of the RESET pin in the case.



Preparation 4 Solder the power wires to the board

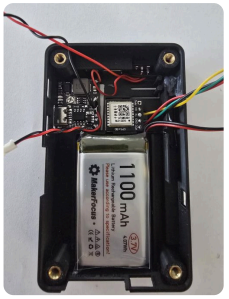


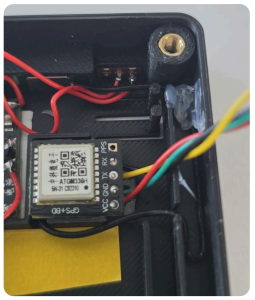
Use two of the three solder pads in the middle of the board: the negative wire to S1 (GND), the positive wire to S3 (5 V). Route the wires flat along the board.

In the photos, the battery and board connections are made with 2-pin 1.25 mm connectors; this is optional. Soldering the wires directly takes up less space. If you use connectors, check the polarity on both sides with a multimeter.

Assembly steps

Step 1 Fit the nuts

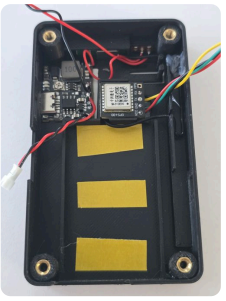




Press the four hex standoff nuts into the pockets in the corner posts. The top of each nut must be flush with the top of the post; if it sticks out, the board will sit crooked.

If a nut is tight, press it in against a flat surface or heat it with a soldering iron for 2–3 seconds. If it is loose, put a drop of glue in the pocket.

Step 2 Apply double-sided tape

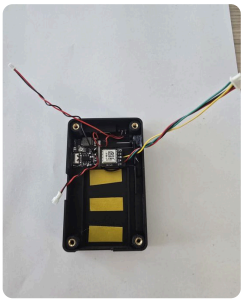


Stick three pieces of thin double-sided tape on the floor of the battery pocket, and one piece each in the GPS module and charge module pockets.

> The tape in the GPS pocket must not cover the GPS LED hole in the floor (the hole the tweezers point to in the photo).

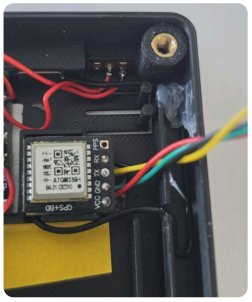
Step 3 Slide the GPS antenna into its pocket

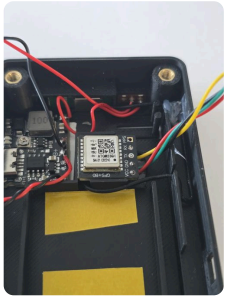




Slide the antenna in from the USB side, into the roofed pocket behind the top wall; the cable exit is on the right (the switch side). If there are strands on the pocket ceiling, remove them first.

Step 4 Route the antenna cable and plug it into the module

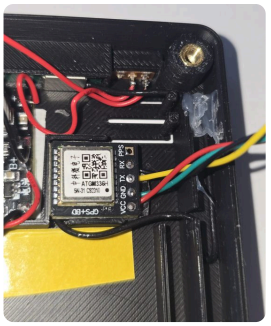


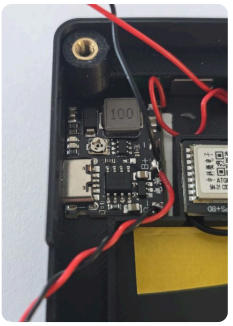


Run the cable down past the post in the top-right corner, clear of the button pins, and on to the GPS pocket. Press the plug onto the gold-colored socket on the GPS module.

Step 5 Place the GPS module

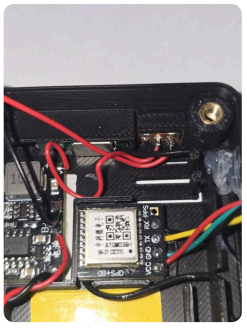
Press the module onto the tape with the labeled side (the one with the QR code) facing you and the LED side toward the floor; the wires point toward the stylus side. Coil the excess antenna cable in a loop along the top edge of the battery pocket.





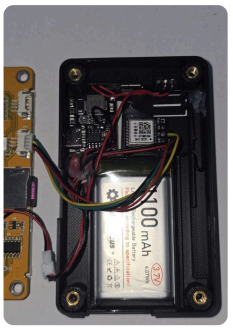
Step 6 Run the GPS cable through the channel

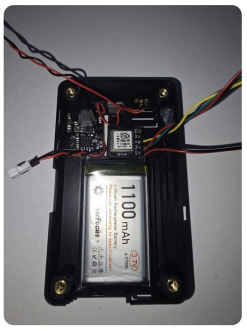
Run the 4-wire cable along the channel up to the CN1 window in the side wall; the plug ends up in front of that window.



Step 7 Place the battery

Press the battery into its pocket with its wires toward the GPS side (top). The antenna cable may run under the battery; it must not pass over the USB sockets.

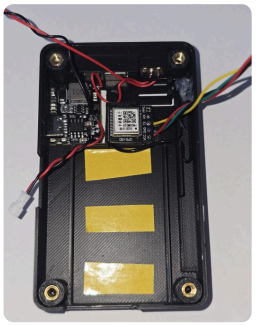


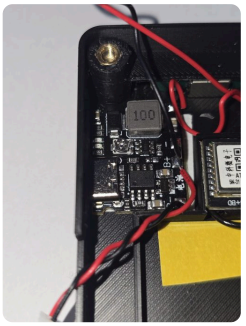


Step 8 Place the charge module and the switch

Press the charge module into its pocket at the top left so that the Type-C socket lines up with the CHG opening and the two charge LEDs line up with the holes in the side wall. Seat the switch in its pocket in the top wall so the lever sticks out through the slot. Connect the battery to the module.

Secure the wires with a few drops of hot glue. Keep the glue off the BOOT / RESET tabs, their pins and the LED holes.

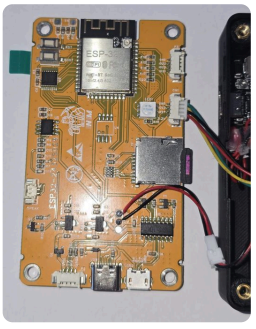




Step 9 Connect the board

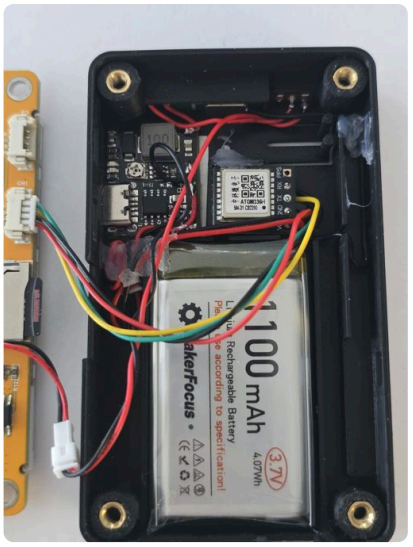
Hold the board next to the case. Connect the power wires (S1 / S3) to the charge module's output and plug the GPS cable into the CN1 socket.

> Test before screwing on the frame: turn the switch on (I, the stylus side). The board should power up and the screen backlight should come on. If the firmware is already installed, the screen also shows an image; if not, see Part 4.



Step 10 Seat the board and screw on the front frame

Lower the board onto the four posts with the screen facing up; make sure no wires are pinched between the board and the posts. Place the front frame over the screen and fit the four M3 × 7 screws from the front. Do not overtighten.



Step 11 Fit the stylus and the SD card

Slide the stylus into the hole in the top wall; its cap sits in the recess at the corner. Insert the SD card through the SD slot in the side wall (push-push slot). The software requires a FAT32-formatted microSD card of up to 32 GB.



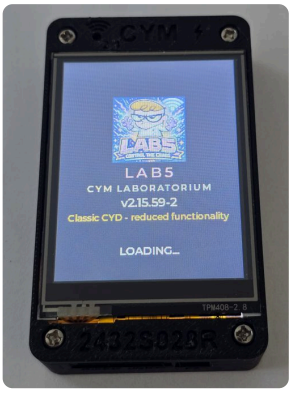


Step 12 Final check

The USB (Type-C and micro) plugs should go into their openings easily; pressing the BOOT and RESET tabs on the back face should click the buttons, and the tabs must not stay pressed on their own. While charging, the charge LEDs show through the holes in the side wall (red while charging, blue when full), and the GPS LED shows through the hole in the back face when GPS is running. The CHG socket is for charging only; it carries no data.

> Keep the switch off (O) while the board's USB socket is connected to a computer or a charger; the board is then powered via USB. You can still charge the battery through the CHG socket at the same time.





Part 4 – Firmware

Flashing the firmware with the Web Flasher

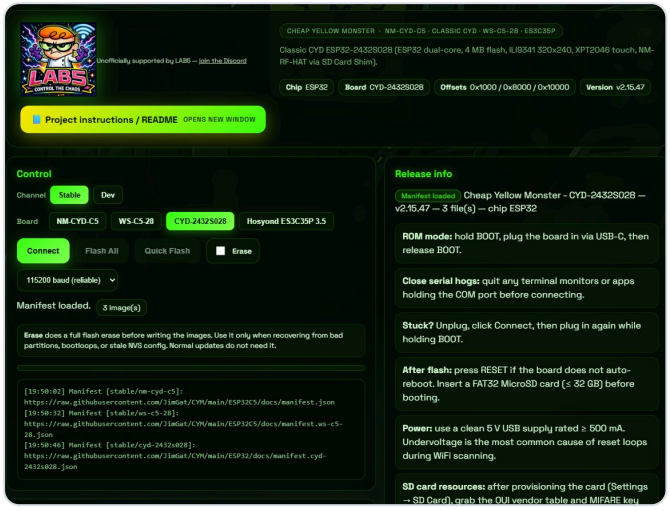
You can flash the firmware directly from the browser, without a USB driver or IDE.

URL: https://jimgat.github.io/CYM/

1. Open jimgat.github.io/CYM/ in Chrome or Edge.
2. Select the "CYD-2432S028" board.
3. Connect the board to the computer via its USB socket and click Connect.
4. Click the "Flash" button and wait for the process to complete.
5. The board restarts automatically; the CYM interface appears on the screen.



When flashing is complete, the CYM main menu appears on the screen.



Web Flasher interface – CYD-2432S028 board selected.

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- **Allowed:** printing and using the case for yourself; sharing the unchanged files with credit to the source and the designers.
- **Not allowed without permission:** selling the printed case or the files, or using them commercially; publishing modified files.
- License text: creativecommons.org/licenses/by-nc-nd/4.0
- The CYM firmware is a separate project under its own license. Board and part names are trademarks of their manufacturers.

> The Web Flasher works only in Chrome and Edge (requires the Web Serial API). Safari and Firefox are not supported.